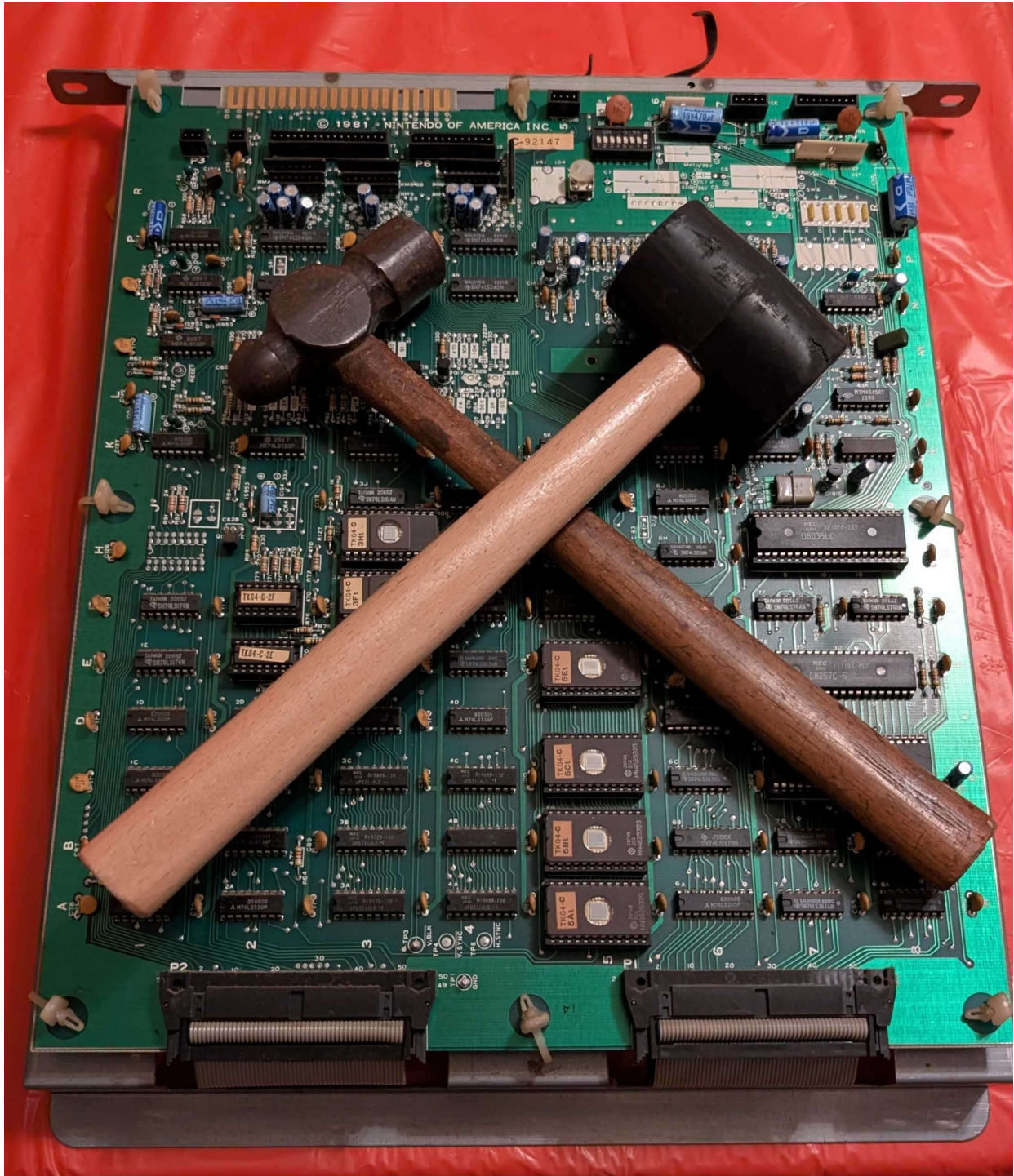


# TKG Test ROM Manual

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# Chapter 1: Overview

## Requirements

This test ROM will not work under every condition. Some things need to be taken into consideration:

- All boards involved must have power, in this case +5V, +12V, and -5V must all be present and within appropriate tolerance, usually within +/- 5%.
- Clocks must be present enough to run. The CPU runs at approximately 3MHz.
- The reset pin and NMI pins must not be held in active. If both are always active, then the ROM will not boot.
- The CPU must be able to access address 0x0000 to 0x0FFF. If the game does not boot it cannot access these ROMs.
- Address and data buffers must be working, if they are not it will not be possible for the program to run.
- Audio OR video must be working. If neither are working, the test ROM will have no way to report failures
- The ROM socket for 5F or 5E (depending on which hardware being tested) must be in good enough condition to make proper contact.

## Compatible Hardware

This test ROM is intended for original TKG hardware. It is compatible with the following hardware:

- Fully compatible
  - TKG2
  - TKG3
  - TKG4
- Partially compatible
  - TRS2 – RAM, ROM, DMA, and NMI tests should all work. However, it is a different game so audio tests will not work. Monitor tests might also not work correctly.

This test ROM will work in the cocktail, cabaret, and upright models of the machine.

## Using the Test ROM

Using the test ROM is very simple. The ROM is burnt onto a 2532 4k EPROM. In order to run the tests you place the EPROM into location 5F for TKG2/TKG3 hardware and 5E TKG4 hardware. Note the notch in the socket, EPROM, and pin 1 markers. After the ROM is inserted reconnect the board back to the cabinet or bench setup for testing.

There are two groups of tests performed by the test ROM: Startup and Runtime. Startup tests are tests that are automatically run when the game boots. This tests RAM, ROM, DMA, NMI, and Audio tests. All tests with the exception of audio tests must pass in order to transition from startup to runtime. Runtime tests are additional tests used for diagnosing other issues with the machine in question. It contains things such as a monitor test and control printouts.

| TKG START-UP TESTS<br>V0.15 22 AUG 2026 |      |         |      | TKG RUN-TIME TESTS<br>V0.15 22 AUG 2026 |     |          |     |
|-----------------------------------------|------|---------|------|-----------------------------------------|-----|----------|-----|
| RAM TESTS                               |      |         |      | CONTROLS TEST                           |     |          |     |
| RAM 0L                                  | PASS | RAM 3L  | PASS | P1 RIGHT                                | OFF | P2 RIGHT | OFF |
| RAM 0H                                  | PASS | RAM 3H  | PASS | P1 LEFT                                 | OFF | P2 LEFT  | OFF |
| RAM 1L                                  | PASS | RAM 4L  | PASS | P1 UP                                   | OFF | P2 UP    | OFF |
| RAM 1H                                  | PASS | RAM 4H  | PASS | P1 DOWN                                 | OFF | P2 DOWN  | OFF |
| RAM 2L                                  | PASS |         |      | P1 JUMP                                 | OFF | P2 JUMP  | OFF |
| RAM 2H                                  | PASS | B. TEST | PASS | P1 START                                | OFF | P2 START | OFF |
|                                         |      |         |      | COIN                                    | OFF |          |     |
| ROM CHECKSUMS                           |      |         |      | DIP SW TEST                             |     |          |     |
| ROM 0                                   | F31E |         |      | 00000001                                |     |          |     |
| ROM 1                                   | 73BA |         |      | ABCDEFGH                                |     |          |     |
| ROM 2                                   | B2BC |         |      |                                         |     |          |     |
| ROM 3                                   | A0F0 |         |      |                                         |     |          |     |
| SYSTEM TESTS                            |      |         |      | SELECT A TEST                           |     |          |     |
| DMA                                     | PASS |         |      | ? PALETTE TEST                          |     | 1        |     |
| NMI                                     | PASS |         |      | SCREEN INVERT                           |     | 0        |     |
|                                         |      |         |      | MONITOR TEST                            |     | 0        |     |
|                                         |      |         |      | SOUND                                   |     | 0        |     |
|                                         |      |         |      | MUSIC                                   |     | 0        |     |
|                                         |      |         |      | BACKGROUND TEST                         |     |          |     |
|                                         |      |         |      | SPRITE TEST                             |     |          |     |
|                                         |      |         |      | RESET                                   |     |          |     |
| AUDIO TESTS                             |      |         |      |                                         |     |          |     |
| SOUND                                   | 6    |         |      |                                         |     |          |     |
| MUSIC                                   | 4    |         |      |                                         |     |          |     |

## Chapter 2: Startup Tests

### RAM Tests

#### RAM Tests

#### RAM Bank Test

#### TRS2 and TKG2 RAM Locations

| Name | Designation      | Type | Start Address | Size   | High Bit Location | Low Bit Location |
|------|------------------|------|---------------|--------|-------------------|------------------|
| RAM0 | Work             | 2114 | 0x6000        | 1k x 4 | C-6L              | C-6H             |
| RAM1 | Work             | 2114 | 0x6400        | 1k x 4 | C-6M              | C-6J             |
| RAM2 | Work             | 2114 | 0x6800        | 1k x 4 | C-6N              | C-6K             |
| RAM3 | Sprite           | 2148 | 0x7000        | 1k x 4 | V-5J              | V-5K             |
| RAM4 | Video/Background | 2114 | 0x7400        | 1k x 4 | V-3K              | V-3J             |

#### TKG3 RAM Locations

| Name | Designation      | Type | Start Address | Size   | High Bit Location | Low Bit Location |
|------|------------------|------|---------------|--------|-------------------|------------------|
| RAM0 | Work             | 2114 | 0x6000        | 1k x 4 | C-6L              | C-6H             |
| RAM1 | Work             | 2114 | 0x6400        | 1k x 4 | C-6M              | C-6J             |
| RAM2 | Work             | 2114 | 0x6800        | 1k x 4 | C-6N              | C-6K             |
| RAM3 | Sprite           | 2148 | 0x7000        | 1k x 4 | V-2L              | V-2M             |
| RAM4 | Video/Background | 2114 | 0x7400        | 1k x 4 | V-5L              | V-5M             |

#### TKG4 RAM Locations

| Name | Designation      | Type | Start Address | Size   | High Bit Location | Low Bit Location |
|------|------------------|------|---------------|--------|-------------------|------------------|
| RAM0 | Work             | 2114 | 0x6000        | 1k x 4 | C-4C              | C-3C             |
| RAM1 | Work             | 2114 | 0x6400        | 1k x 4 | C-4B              | C-3B             |
| RAM2 | Work             | 2114 | 0x6800        | 1k x 4 | C-4A              | C-3A             |
| RAM3 | Sprite           | 2148 | 0x7000        | 1k x 4 | V-6R              | V-6P             |
| RAM4 | Video/Background | 2114 | 0x7400        | 1k x 4 | V-2R              | V-2P             |

### ROM Checksums

#### Test Overview

The ROM test performs a 16-bit checksum on the contents of all ROMs. This test does not report the integrity of the ROMs nor does it validate that it is correct for a location. That is up to the user to determine. The exception to this is ROM 0, which is where the test ROM is located.

## Test ROM Integrity Test

As part of the ROM test, ROM 0 will be validated. ROM 0 is the test ROM itself. Its own checksum is appended to the footer as part of a post-build script. It will compare the measured checksum with what is expected. If there is any corruption of the test ROM or if there is issues with accessing say the upper addresses, it will report a failure by printing “Integrity Fail”. In this case, the test ROM may not be trustworthy and could run into issues during execution.



### ROM Locations

| ROM ID | ROM Type | ROM Size | Start Address | TKG2/TKG3 Location | TKG4 Location |
|--------|----------|----------|---------------|--------------------|---------------|
| ROM 0  | 2532     | 4k       | 0x0000        | 5F                 | 5E            |
| ROM 1  | 2532     | 4k       | 0x1000        | 5G                 | 5C            |
| ROM 2  | 2532     | 4k       | 0x2000        | 5H                 | 5B            |
| ROM 3  | 2532     | 4k       | 0x3000        | 5K                 | 5A            |

### ROM Checksum Table

| ROM ID | US Set 1 | US Set 1 Hard | US Set 2 | Japan Set 1 | Japan Set 2 | Japan Set 3 |
|--------|----------|---------------|----------|-------------|-------------|-------------|
| ROM 0  |          |               |          |             |             |             |
| ROM 1  | 73BA     |               |          |             |             |             |
| ROM 2  | B2BC     |               |          |             |             |             |
| ROM 3  | A0F0     |               |          |             |             |             |

## DMA Test

TKG hardware relies heavily on DMA to transfer data from work RAM to sprite RAM. During gameplay it is triggered during the NMI routine. The reason for this is because this is when vertical blanking (also known as VBlank) is occurring, meaning that nothing is being drawn on screen. VBlank occurs sixty times per second. On TKG hardware, an 8257 DMA controller is used to perform these

operations as well as some supporting circuitry. On TKG4 hardware, that circuitry includes the 74LS373 octal latch at 6B, the 74ls08 quad and-gate at 5F, and the 74LS125 buffer at 6E. The 8257 derives its clock from the 1H signal (~3MHz). Unlike the CPU clock however, it goes through an inverter (74LS04 at 6D). Chip select circuitry is also involved. Refer to the schematics to find out more.

The DMA test works by initiating a transfer from the ranges \$6900-\$6A80 to \$7000-\$7180. This test mimics what would be observed in real game conditions. This test is run three times with different byte values. After the DMA transfer is kicked off, the CPU reads back the data that should have been copied over and compares it to what is in the work RAM mirror. If all bytes do not match, then the test is marked as “fail” The DMA test can only occur if RAM2 and RAM3 are functional. If either of these RAM banks failed its RAM test, the DMA test will not process the results and instead print N.A.

A severely failing 8257 DMA controller can cause the board to be unable to boot in rare cases. The DMA controller has direct access to the databus shared with the CPU. For the purposes of the test ROM, you should be able to remove the DMA controller to be able to perform startup tests. However, it will not proceed past the startup tests due to the DMA failure. Be aware that the DMA controller is not always in an IC socket.

## **NMI Test**

## **Audio Tests**

### **Sound Test**

### **Music Test**

## **Transition to Runtime**

## **Chapter 3: Runtime Tests**

**Controls Test**

**DIP Switch Test**

**Palette Test**

**Screen Invert**

**Monitor Test**

**Sound Test**

**Music Test**

**Background Test**

**Sprite Test**

**Reset**

## Chapter 4: Test Architectures

# Abbreviations

**CPU:** Central Processing Unit

**EPROM:** Erasable Programable Read Only Memory

**IC:** Integrated Circuit

**N.A.:** Not Applicable

**NMI:** Non-Maskable Interrupt

**RAM:** Random Access Memory

**ROM:** Read Only Memory